1 Baseline Replication

For the baseline replication, one core model and one core dataset from the original study were selected:

- **Model:** ALBERT-v2
- **Dataset:** Expanded Multi-Grain Stereotype Dataset (EMGSD)

ALBERT-v2 was chosen as the base model due to its parameter-efficient design and shared encoder layers, which support scalable deployment in real-world moderation systems [3].

1.1 Data Preprocessing and Splitting

The EMGSD dataset was preprocessed to create binary classification labels. Instances with labels beginning with "stereotype_" were mapped to class 1 (Stereotype), while all other instances were mapped to class 0 (Non-stereotype). The dataset was split into training (64%), validation (16%), and test (20%) sets using stratified sampling to maintain class distribution across splits (random seed = 42).

1.2 Training Configuration

Table 1: Hyperparameters for ALBERT-v2 fine-tuning on EMGSD

Hyperparameter	**Value**
Base model	`albert-base-v2`
Max sequence length	128
Batch size (train / eval)	32 / 32
Number of epochs	2
Learning rate	2×10^{-5}
Weight decay	0.01
Loss function	Cross-entropy (via `AutoModelForSequenceClassification`)
Optimizer / scheduler	HuggingFace defaults for `TrainingArguments` (`AdamW`-style optimiser)

All experiments were conducted using the Hugging Face Transformers library (version 4.x) with PyTorch backend. The baseline implementation was reproduced using publicly released code and configurations. Data preprocessing steps, model architecture, and training procedures were aligned as closely as possible with the original experimental setup. Minor deviations were necessary due to differences in data availability and computational constraints.

1.3 Results

The replicated baseline model achieved strong performance on the EMGSD test set, as shown in Table 2. The model obtained a macro F1-score of 0.819 and an overall accuracy of 0.839, successfully demonstrating the reproducibility of the original study's methodology.

Table 2: Performance of the replicated baseline model (ALBERT-v2 on EMGSD)

Class	**Precision**	**Recall**	**F1-score**
0 (Non-stereotype)	0.870	0.888	0.879
1 (Stereotype)	0.775	0.744	0.759
Macro Avg	0.823	0.816	0.819
Overall Accuracy: 0.839			

Table 3: F1-score compared with the original study

Model	**Dataset**	**F1-score (Macro Avg)**
ALBERT-v2 (Original)	EMGSD	0.819
ALBERT-v2 (Replicated)	EMGSD	0.823

2 GBV Dataset Construction

To adapt the HEARTS framework to a gender-based violence (GBV) detection context, a new dataset was constructed using the *Jigsaw Unintended Bias in Toxicity Classification* dataset. The dataset was originally designed to evaluate unintended bias by providing identity-related annotations [1]. In this study, these annotations were used to construct a subset of comments specifically related to women, enabling the development of a GBV-focused classification dataset. This approach allows the model to focus on detecting hostility directed toward women while distinguishing it from neutral identity-related discourse.

A GBV-focused subset was constructed by filtering comments that explicitly reference women or girls, using a combination of identity-based annotations and keyword-driven rules. A binary label was subsequently derived, where a comment is classified as *hostile* if it exhibits both high overall toxicity and at least one strong harm-related subtype (e.g., insult, threat, or identity attack), with a threshold greater than 0.3.

Manual inspection shows that, although the filtering procedure substantially increases the proportion of comments about women, a minority of retained instances are not explicitly gender-based or GBV-related. This reflects noise in the underlying toxicity annotations and the limitations of rule-based filters.

This filtering process resulted in a dataset comprising 86,719 comments, including 10,339 hostile and 76,380 non-hostile instances. The resulting class distribution reflects a realistic yet imbalanced setting commonly observed in real-world online moderation tasks.

3 Exploratory Data Analysis

Exploratory Data Analysis (EDA) was conducted to examine the characteristics of the constructed GBV-focused dataset and to validate the effectiveness of the filtering strategy.

- **Text Length Distribution:** Analysis of comment lengths reveals a right-skewed distribution, with the majority of comments being relatively short and a smaller proportion consisting of longer, more descriptive posts. This suggests substantial variability in linguistic expression and supports the use of transformer-based models capable of capturing both short and long-range contextual information.

- **Word Cloud:** The word cloud visualization highlights frequently occurring terms within the dataset, including identity-related references (e.g., *woman*, *men*, *man*) and emotionally charged language. The prominence of such terms indicates that the dataset captures meaningful gender-related discourse.

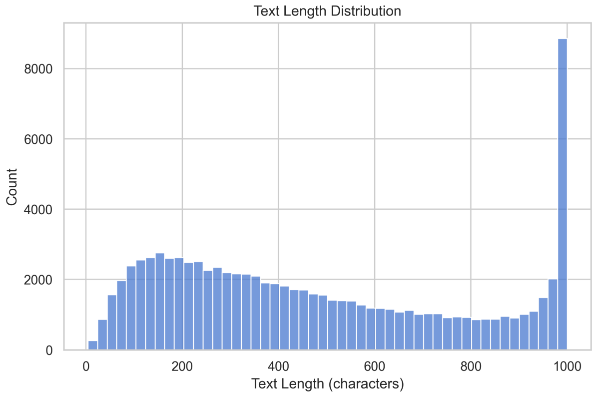

Figure 1: Text length distribution

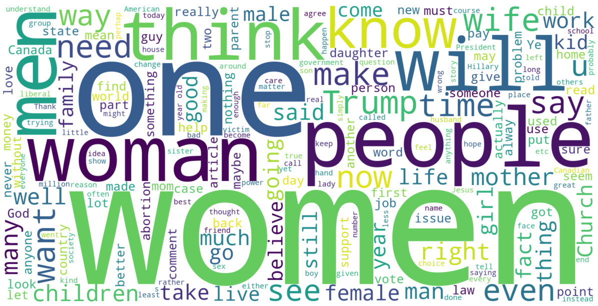

Figure 2: Word Cloud

4 Experimental Setup

4.1 Model and Tokenisation

For the GBV classification task, I fine-tuned the pretrained `albert-base-v2` model using the HuggingFace `transformers` library. The corresponding `AutoTokenizer` was used for text preprocessing. Comments were tokenised with:

- maximum sequence length = 128 tokens (`max_length = 128`),
- truncation of longer comments,
- padding to fixed length (`padding = "max_length"`).

4.2 Data Splits

The Jigsaw GBV dataset contains comments with binary labels (*non-hostile* / *hostile*). The dataset was split into training (64%), validation (16%), and test (20%) sets using stratified sampling to maintain class distribution across splits (random seed = 42).

4.3 Training Configuration

Fine-tuning was performed using the `Trainer` API with the following hyperparameters:

Table 4: Hyperparameters for ALBERT-v2 fine-tuning on the GBV dataset

Hyperparameter	**Value**
Base model	`albert-base-v2`
Max sequence length	128
Batch size (train / eval)	32 / 32
Number of epochs	3
Learning rate	2×10^{-5}
Weight decay	0.01
Loss function	Cross-entropy (via `AutoModelForSequenceClassification`)
Optimizer / scheduler	HuggingFace defaults for `TrainingArguments` (`AdamW`-style optimiser)

5 Performance Evaluation

5.1 Classification Metrics

During training, a held-out validation set was used for intermediate evaluation. After training, the best model was saved and evaluated on the test set. The following metrics were computed:

Table 5: Performance of ALBERT-v2 on the GBV dataset

Class	**Precision**	**Recall**	**F1-score**
0 (Non-hostile)	0.938	0.966	0.952
1 (Hostile)	0.676	0.525	0.591
Macro Avg	0.807	0.745	0.771
Overall Accuracy: 0.913			

The adapted model achieved an accuracy of 91.3% and a macro-averaged F1 score of 0.771. Performance was notably stronger on the non-hostile class, with high precision and recall, while recall for the hostile class was lower. This discrepancy reflects the inherent class imbalance and the linguistic diversity of hostile expressions.

Compared to the baseline HEARTS results in Table 1, the adapted model demonstrates comparable overall performance while operating in a more challenging, context-specific classification setting. Although overall accuracy remains high, macro-averaged recall and F1 scores are lower. This reflects the increased difficulty of detecting implicit and context-dependent hostility in online GBV, as opposed to more surface-level stereotypical language.

5.2 Model Interpretability & Error Analysis

Table 6: LIME explanations for representative prediction cases

Text Instance	Predicted	Actual	Top Token Rankings (LIME)
RAPO in the White House. Terrorists and fanatics at your local Planned Parenthood. Get your Glocks women.	1	0	“Terrorists”: 0.692, “your”: 0.159, “RAPO”: 0.123, “fanatics”: 0.099, “White”: 0.087, “Glocks”: 0.078
You must be male. Your favorite politicians certainly want dominion over my body and all other women’s. So I guess your idea of “slavery” and “tyranny” is sliiiiiggghhhhtttllly…	0	1	“silly”: 0.391, “idea”: −0.106, “and”: −0.085, “liberty”: −0.069, “favorite”: −0.066, “death”: 0.063
The “broad” is the dead kid’s mother you insensitive idiot.	1	1	“idiot”: 0.889, “insensitive”: 0.038, “broad”: −0.036, “dead”: 0.018, “you”: 0.012, “mother”: −0.010

Model interpretability was examined using Local Interpretable Model-agnostic Explanations (LIME) [4]. Analysis of **true positive** examples shows that explicit insults and overtly hostile language strongly influence correct hostile predictions. **False positive** cases indicate that the model can overemphasise emotionally charged or negative terms that are not necessarily gender-targeted, leading to misclassification of non-hostile comments. In contrast, **false negative** cases often involve indirect, contextual, or descriptive language that lacks explicit hostility markers. Together, these findings highlight the model’s sensitivity to explicit cues while revealing limitations in detecting subtle or implicit forms of gender-based hostility.

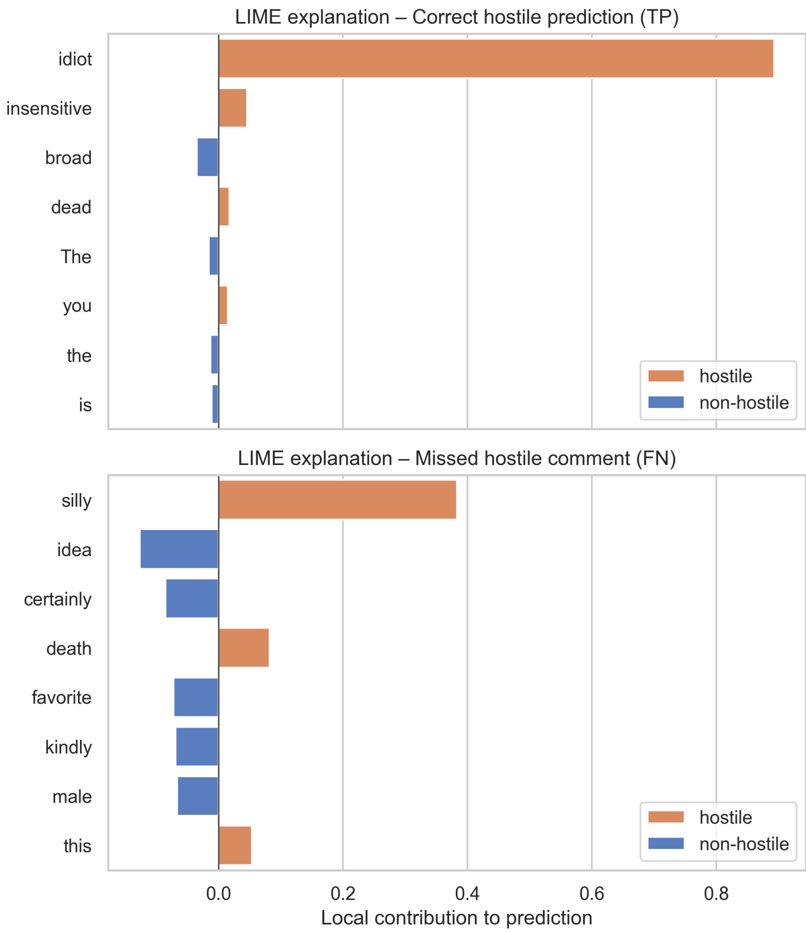

Figure 3: LIME explanations for a true positive and a false negative prediction

5.3 Lexical Ablation Analysis

A lexical masking ablation was conducted to assess the model's reliance on explicit abusive terms [2]. High-frequency tokens were extracted from true positive hostile predictions, and the top 80 tokens were selected as lexical cues. The experiment was conducted on a randomly sampled subset of 2,000 posts. These tokens were replaced with a neutral placeholder (***), while preserving the remaining text. The trained model was evaluated on both original and masked inputs without retraining.

Table 7: Lexical masking ablation results on the GBV test set

Metric	**Original Input**	**Masked Input**
Macro F1-score	0.777	0.727
Hostile Recall	0.552	0.418
Mean Hostile Probability Drop	–	0.119
Median Hostile Probability Drop	–	0.031
Hostile $\rightarrow$ Non-hostile Flip Rate	–	0.033

The lexical masking ablation provides clear evidence that the model's predictions are strongly influenced by explicit lexical cues. When high-frequency abusive terms are masked, performance degrades consistently across multiple metrics. In particular, the macro-averaged F1 score decreases from 0.777 to 0.727, and recall for the hostile class drops substantially from 0.552 to

0.418. This indicates that a significant portion of the model's ability to identify hostile content relies on the presence of overtly abusive language.

Moreover, the observed reduction in predicted hostile probabilities further supports this interpretation. For samples that are truly hostile, masking explicit lexical cues leads to a mean probability decrease of approximately 0.12, suggesting that the model's confidence in hostile predictions is systematically weakened even when the underlying semantic context remains unchanged. The flip rate from hostile to non-hostile predictions (3.3%) under masking highlights that, for a subset of cases, explicit lexical cues are decisive for correct classification.

6 Ethical Considerations

Automated detection of hostile language raises important ethical concerns, including the risk of false positives suppressing legitimate speech and false negatives allowing harmful content to persist. Dataset bias, annotation subjectivity, and uneven representation of identity groups can further exacerbate these risks. This project mitigates some concerns through transparency, using explainability tools to expose model decision logic.

7 Limitations

Several limitations remain in this study. First, a limitation stems from the dataset construction process. Deriving a binary GBV label from continuous toxicity scores and harm-related subtypes may oversimplify the heterogeneous nature of online GBV. Consequently, implicit or indirect expressions of hostility toward women are less likely to be captured than overtly abusive language, potentially biasing model training toward more explicit forms of harm.

This is further reflected in the model's behavior, as evidenced by the LIME explanations and the lexical masking ablation analysis, which indicate a strong reliance on explicit abusive terms. Future work should focus on refining the model's local decision boundaries to ensure the model distinguishes between abusive terms and implicit hostility toward women [2].

Second, the model operates at the level of isolated comments and does not incorporate conversational or temporal context. As a result, forms of GBV that emerge through repeated interactions, escalation, or cumulative harm may not be adequately captured, limiting the model's applicability to real-world moderation scenarios.

8 Scalability and Sustainability

The use of ALBERT significantly reduces memory requirements and computational cost compared to larger transformer models, supporting scalability in real-world moderation systems. Once trained, the model can efficiently process large volumes of text, making it suitable for platform-level deployment.

From a sustainability perspective, parameter-efficient architectures reduce energy consumption during training and inference. Nevertheless, regular retraining is required to adapt to evolving language use, which carries ongoing environmental and computational costs.

From a policy and deployment perspective, automated GBV detection systems should be integrated within human-in-the-loop moderation pipelines rather than used as fully autonomous enforcement mechanisms. Platform-level deployment should prioritize transparency, appeal mechanisms for affected users, and regular auditing of model performance to mitigate bias and unintended harm.

References

[1] Daniel Borkan, Lucas Dixon, Jeffrey Jerebschi, Geoff Cain, and Nithum Vasserman. Nuanced metrics for measuring unintended bias with real data for text classification. In *Companion Proceedings of the 2019 World Wide Web Conference*, pages 491–500, 2019.

[2] Matt Gardner, Yoav Artzi, Victoria Basmova, Jonathan Berant, Ben Bogin, Sihao Chen, Pradeep Dasigi, Dheeru Dua, Yanai Elazar, Ananth Gottumukkala, et al. Evaluating models' local decision boundaries via contrast sets. *arXiv preprint arXiv:2004.02709*, 2020.

[3] Zhenzhong Lan, Mingda Chen, Sebastian Goodman, Kevin Gimpel, Piyush Sharma, and Radu Soricut. Albert: A lite bert for self-supervised learning of language representations. In *International Conference on Learning Representations*, 2019.

[4] Marco Tulio Ribeiro, Sameer Singh, and Carlos Guestrin. "why should i trust you?": Explaining the predictions of any classifier. In *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, pages 1135–1144, 2016.